

Braid Arrangements for Concurrent Programs

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Abstract

Precubical sets are a standard tool in the study of concurrent programs. A recent paper by Paliga and Ziemiański showed that the fundamental group for the path space on the final precubical set was exactly is the Artin braid group. However, they also showed the induced braid subgroups for euclidean sets trivialized. In the world of hyperplane arrangements, this kind of trivialization can be resolved by working in the “complexified” space. We use this same approach, to build a new, complexified version of the path space, Ch^* . This is homotopy equivalent to the Salvetti Complex of the braid arrangement in for the cube, but defined over any precubical set. By Salvetti’s theorem, then, $\pi_1|\text{Ch}^*(\square^n)|$ is the pure braid group. This gives us a new braid invariant for concurrent programs.

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1 Introduction

This work is in 3 parts. First we’ll recap the relevant material in brief. Second, we will perform the construction of the “complexified” path space. Third, we will study some of its properties.

To fix notation, let Cube be the box category: its objects are the cubes $\square^n$ for $n \geq 0$, and its morphisms are the face embeddings (composites of the coface maps δ_i^ϵ). A *precubical set* is a presheaf on Cube ; write $\square$ for the category of precubical sets. Concretely, a precubical set K has, for each $n \geq 0$, a set $K[n]$ of n -cells, together with face maps

$$d_i^\epsilon : K[n] \longrightarrow K[n-1] \quad (1 \leq i \leq n, \epsilon \in \{0, 1\}),$$

where i names a direction and ϵ a side, subject to the precubical identities $d_i^\epsilon d_j^\eta = d_{j-1}^\eta d_i^\epsilon$ for $i < j$. We write $\square^n$ for the standard n -cube, the presheaf represented by $\square^n \in \text{Cube}$.

We usually work with precubical sets carrying designated start and end vertices. A *bipointed* precubical set is a precubical set K with two chosen vertices $\text{init}_K, \text{fin}_K \in K[0]$, and morphisms are required to preserve them; write $\square_*^*$ for the resulting category. Each cube $\square^n$ is bipointed by its initial vertex $(0, \dots, 0)$ and final vertex $(1, \dots, 1)$.

2 Standard build

We recall three constructions on $\square_*^*$ from Paliga–Ziemiański.

The first is the **wedge** $\vee$. Given $P, Q \in \square_*^*$, the wedge $P \vee Q$ is the pushout gluing P 's final vertex to Q 's initial vertex:

$$\begin{array}{ccc} \square^0 & \xrightarrow{\text{init}_Q} & Q \\ \text{fin}_P \downarrow & \lrcorner & \downarrow \\ P & \longrightarrow & P \vee Q \end{array}$$

The wedge $P \vee Q$ is bipointed by init_P and fin_Q , and $(\square_*^*, \vee, \square^0)$ is monoidal with unit the point $\square^0$. The **serial wedge** of a list of naturals is

$$\bigvee[a_1, \dots, a_n] := \square^{a_1} \vee \dots \vee \square^{a_n},$$

with $\bigvee[] = \square^0$.

The second construction is the category of **cube chains**. For $K \in \square_*^*$, let $\text{Ch}(K)$ be the full subcategory of the slice $\square_*^*/K$ spanned by the serial wedges: an object is a bipointed map $\bigvee a \rightarrow K$, and a morphism $(\bigvee a \rightarrow K) \rightarrow (\bigvee b \rightarrow K)$ is a bipointed map $\bigvee a \rightarrow \bigvee b$ commuting over K . Such a morphism runs from a finer chain to a coarser one; for instance the two-edge path $\bigvee[1, 1] \rightarrow \square^2$ carries a morphism to the identity chain $\square^2 \xrightarrow{\text{id}} \square^2$.

The third construction is **refinement**. A *chain* in K is a finite sequence $[c_1, \dots, c_l]$ of cells of K , each of dimension at least 1, such that $\text{fin}(c_i) = \text{init}(c_{i+1})$ for all i , with $\text{init}(c_1) = \text{init}_K$ and $\text{fin}(c_l) = \text{fin}_K$; here $\text{init}(c)$ and $\text{fin}(c)$ denote the images under $c : \square^{\dim c} \rightarrow K$ of the initial and final vertices of the cube. A *refinement* replaces a cell c_i by a chain of the cube $\square^{\dim c_i}$, post-composed with c_i . Let $\text{Refine}(K)$ be the category of chains, with morphisms generated by refinements and oriented as in $\text{Ch}(K)$ (finer to coarser).

Finally, two properties of K from the Configuration Spaces paper recur. We say K *admits altitude* if there is a function $\text{alt} : K[0] \rightarrow \mathbb{Z}$ with $\text{alt}(\text{fin}(e)) = \text{alt}(\text{init}(e)) + 1$ for every edge $e \in K[1]$. We say K is *non-self-linked* (NSL) if every map $\square^n \rightarrow K$ is mono.

A few things to note

Lemma 2.1. *The $\vee$ operator is a monoidal product on $\square_*^*$.*

Lemma 2.2. *$\text{Ch} : \square_*^* \Rightarrow \text{Cat}$ is a lax monoidal functor from $(\square_*^*, \vee)$ to $(\text{Cat}, \times)$*

Lemma 2.3. *If K admits altitude and is NSL, then $\text{Ch}(K)$ and $\text{Refine}(K)$ are equivalent, so Ch is strong monoidal there.*

Lemma 2.4. *Serial wedges admit altitudes, and are NSL*

This is all the foundational material we need.

3 Wedge Machinery

In an effort to keep our proofs better factored, we establish some results here that let us extend construction on Cube to constructions on $\square_*^*$.

Lemma 3.1. *Any functor $P : \text{Cube} \Rightarrow \text{Cat}$ extends, by left Kan extension along $\text{Cube} \hookrightarrow \square$, to a functor $E(P) : \square_*^* \Rightarrow \text{Cat}$. If $P(\square^0) = \emptyset$, then $E(P)$ is strong monoidal*

$$E(P) : (\square_*^*, \vee, \square^0) \Rightarrow (\text{Cat}, \sqcup, \emptyset),$$

and for every serial wedge

$$E(P)(\bigvee a) \simeq \coprod_i P(\square^{a_i}).$$

Proof. As a pointwise left Kan extension (a coend over Cube), $E(P)$ is cocontinuous, so it sends the wedge pushout $A \leftarrow \square^0 \rightarrow B$ to $E(P)(A \vee B) = E(P)(A) \sqcup_{P(\square^0)} E(P)(B)$. When $P(\square^0) = \emptyset$ this pushout is the coproduct, so the comparison map is an isomorphism; the formula for $\bigvee a$ follows by induction on the number of summands. $\square$

Lemma 3.2. *Any functor $P : \text{Cube}^{op} \Rightarrow \text{Cat}$ extends, by right Kan extension along $\text{Cube} \hookrightarrow \square$, to a Cat -valued presheaf $E(P) : \square_*^{*op} \Rightarrow \text{Cat}$. If $P(\square^0)$ is terminal, then $E(P)$ is strong monoidal*

$$E(P) : (\square_*^{*op}, \vee, \square^0) \Rightarrow (\text{Cat}, \times, \mathbf{1}),$$

and for every serial wedge

$$E(P)(\bigvee a) \simeq \prod_i P(\square^{a_i}).$$

Proof. Dually, $E(P)$ is a pointwise right Kan extension, hence continuous, and being contravariant it sends the wedge pushout to the pullback $E(P)(A \vee B) = E(P)(A) \times_{P(\square^0)} E(P)(B)$. When $P(\square^0)$ is terminal this pullback is the product, so the comparison is an isomorphism; the formula for $\bigvee a$ follows by induction. $\square$

Together, these give us tools for lifting and decomposing cube maps to tensor maps.

So let us define the cases we care about

Definition 3.3. Let $\text{Coord} : \text{Cube} \Rightarrow \text{Cat}$ send $\square^n \mapsto [n]$, and a map $f : \square^n \rightarrow \square^m$ is a face inclusion which induces a $[n] \rightarrow [m]$. This is a functor. And $[0]$ is empty, so it satisfies the lifting condition. This gives us $\text{coordMap} : (\bigvee a \rightarrow \bigvee b) \rightarrow (\Sigma_i [a_i] \rightarrow \Sigma_j [b_j])$ directly as a monoidal functor. This tells us how coordinate positions in $\bigvee a$ translate to positions in $\bigvee b$.

Lemma 3.4. *For any $f : \bigvee a \rightarrow \bigvee b$, the function $\text{coordMap}(f)$ is bijective.*

Proof. We can by induction and monoidality, reduce this to just the case $\bigvee a \rightarrow \square^m$. Let's start with injectivity. We can induct on the size of a . The base case is trivial. So we can consider $(\bigvee b) \vee a \rightarrow \square^m$, and coordMap induces an injection on $\bigvee b \rightarrow [m]$. Then every vertex in the image of a is reachable from the final vertex of the image of $\bigvee b$. Which means the their coordinates greater elementwise. Since the image of the induced $[a] \rightarrow [m]$ is only the coordinates that flipped, the images are disjoint. And since $[a] \rightarrow [m]$ is injective, we can union the two functions.

For surjectivity, observe that the serial wedges and cubes admit altitude, and so the altitudes of their final points must agree. So every coordinate in m is in the image. So coordMap is indeed a bijection. The $\square$

4 The Complexification Construction

The construction starts by observing that one dimensional serial wedges behave like topes, or chambers, from the hyperplane configuration world. That is, they are the maximally refined elements. Let's start by reviewing the story on the braid arrangement side.

Following Paris, but specifically for the braid arrangement. The $n - 1$ braid arrangement is a family of hyperplanes in $\mathbb{R}^n$ of $\{x_i = x_j\}$ for $0 \leq i < j \leq n$. The idea is to give each plane an orientation and compute $x \mapsto x_i - x_j$. So that we can break up $\mathbb{R}^n$ into faces represented as functions

$$\mathbf{Face}(\mathbf{n}) := [n] \rightarrow \{-1, 0, 1\}$$

These are called the covectors. We can order covectors

$$x \leq y := \forall i, x(i) = 0 \vee x(i) = y(i)$$

which gives an order on $\mathbf{Face}(\mathbf{n})$. Maximal elements of this order are covectors that are never 0, which are precisely the complement of the hyperplanes. We call these maximal elements **topes**.

The main issue with **Face** is that it is contractible, so it's hard to reason about. The goal of Salvetti's theorem is to build a structure on top of the same covectors that captures essentially the same information, but with a richer structure. Repeat the process, but with $\mathbb{C}$ instead of $\mathbb{R}$. We end up with the Salvetti Poset Sal .

$$\text{Sal}(n) = \{(F, C) \in \mathbf{Face}(\mathbf{n}) \times \mathbf{Face} \mid C \text{ is a tope} \wedge F \leq C\}$$

We need one more definition: the composition of two faces

$$F \circ G := i \mapsto \text{if } F(i) = 0 \text{ then } Y(i) \text{ else } X(i)$$

For a tope C and face F , we define C_F as the

$$(F, C) \preceq (F', C') := F \leq F' \wedge C' = F' \circ C$$

This poset induces a complex, and Salvetti's theorem tells us that it is isomorphic to the configuration space of n points. So, its fundamental group is pure braids. Intuitively, we build a covering-like structure, where we pair each face with a direction annotation, indicated by a compatible tope. So when we paths cross through faces, they can cause orientation changes. These changes are what the braid words represent. And in fact, this direction information is what the Configuration Spaces paper retains as well, essentially breaking every tie at a face in every possible way.

This construction abstracts a bit, in a way that will make it convenient to construct for us. In particular, there's a functor $\text{Tope} : \text{Face} \Rightarrow \{X \in \text{Face}, X \text{ is a tope}\}$ where each face maps to its neighboring topes. And functions between faces induce orientation-preserving maps of topes. Then the salvetti poset is $\int \text{Tope}$, the Grothendieck construction.

To rebuild this generically on the precubical side, let's start with the corresponding definition of tope.

Definition 4.1. A serial wedge is **one dimensional** if all its entries are 1

Definition 4.2. Let $\text{Run}(K)$ be the full subcategory of $\text{Ch}(K)$ such that the domains are one dimensional serial wedges.

Lemma 4.3. *Being one dimensional is preserved under $\vee$, so Run is a monoidal functor. And in particular $\text{List}\mathbb{N} \Rightarrow \text{Run}(\vee \cdot)$ is a lax monoidal functor.*

Now, the morphisms we want to build are a bit unusual, because they are built from degeneracies, which are not morphisms in Precubical. That is, we want to build a map from an op arrow in $\text{Ch}(K)$ to runs that mirrors the order from Salvetti.

$$(\bigvee a \longrightarrow \bigvee b) \rightarrow (\text{Run}(\bigvee b) \rightarrow \text{Run}(\bigvee a))$$

To achieve this, let $\text{Edge}(K)$ be the chains in $\text{Refine}(K)$ with each cell 1 dimensional. Then $\text{Run}(K) = \text{Edge}(K)$.

Now, given a face embedding $(f : \square^n \longrightarrow \square^m)$, it defines a unique projection of cells of m to cells of n , non-increasing on dimension. Call it $\hat{f}$. So we define

$$\rho : (f : \square^n \longrightarrow \square^m) \rightarrow (\text{Run}[\square^m] \rightarrow (\text{Run}[\square^n]))$$

via

$$\rho' : (f : \square^n \longrightarrow \square^m) \rightarrow (\text{Refine}_1[\square^m] \rightarrow (\text{Refine}_1[\square^n]))$$

by

$$[c_1, \dots, c_k] \mapsto [\hat{f}(c_1), \dots, \hat{f}(c_k)]$$

Note that this is indeed a one dimensional chain in $\square^n$ because $\hat{f}$ never increases dimension, and every projection respects initial/final points.

Now, ρ is a nice. In particular,

Lemma 4.4. $\rho : \text{Cube}^{\text{op}} \Rightarrow \text{Set}$ is a functor, and $\rho(\square^0)$ is a singleton

Proof. The proof of functorality mostly just exploits the fact that cubes are NSL and admit altitudes. So $\text{Refine}_1(\square^n) \simeq \text{Run}(\square^n)$ as sets. First, ρ preserves identities, as the projection for with the identity embedding is also the identity. Second, if $f : \square^n \rightarrow \square^m$ and $g : \square^m \rightarrow \square^n$ are face embeddings, then their corresponding projections respect composition.

For the singleton, observe that there $\rho(\square^0) = \text{Run}[\square^0]$, but the only one dimensional serial wedge which maps into $\square^0$ is the the empty one. So indeed $\rho(\square^0)$ is a singleton. $\square$

That means two things. Firstly, it means ρ is itself, a precubical set. Second, it means our lifting machinery sends ρ to $\hat{\rho} : \square_*^* \Rightarrow \text{Cat}$. Lastly, let $\square^\vee$ be the full subcategory of serial wedges in $\square_*^*$. There is also a forgetful functor $W(K) : \text{Ch}(K) \Rightarrow \square^\vee$ sending $\bigvee a \rightarrow K \mapsto \bigvee a$. Then we can define our key functor

$$\begin{aligned} \text{Lines}(K) &:= E(\rho) \circ W(K) \\ \text{Ch}^*(K) &:= \int \text{Lines} \end{aligned}$$

Defining Ch^* abstractly has some conveniences later. In particular, since E is functorial, we can replace ρ with an equivalent presheaf to produce an equivalence of $\text{Ch}^*(\square^n) \simeq \text{Sal}(n)$. But it's worth unpacking the definition of Ch^* to have a more concrete presentation of it.

Objects of $\text{Ch}^*(K)$ are pairs

$$(\bigvee a \longrightarrow K, x : \text{Run}[\bigvee a]), \text{ I.E. } \bigvee[1, \dots, 1] \longrightarrow \bigvee a \longrightarrow K$$

The intuition here matches the Salvetti story precisely, we have a wedge map into K , coupled with orientation info about it. Morphisms are a little uglier. A morphism

$$(\bigvee a \longrightarrow K, x : \text{Run}[\bigvee a]) \longrightarrow (\bigvee b \longrightarrow K, y : \text{Run}[\bigvee b])$$

unpacks to

1. A morphism $f : (\bigvee a \longrightarrow K) \longrightarrow (\bigvee b \longrightarrow K)$ I.E. a commuting $f' : \bigvee a \longrightarrow \bigvee b$.
2. A $\phi : \text{Lines}(F)(x) \rightarrow y$, in other words $\phi : E(\rho)(f')(x) \rightarrow y$. This means we are taking a run x , passing it through the projection maps, and comparing it with y . The key observation here is that these morphisms can be represented by permutations. A map in $\text{Run}[\bigvee b]$ is a map $\bigvee[1, \dots, 1] \rightarrow \bigvee[1, \dots, 1]$. Which by the coordinate map, a bijection $[n] \rightarrow [n]$. Let $\sigma(f)$ be that permutation.

Below is the diagram for this story. The solid arrows are morphisms of $\text{Ch}^*(K)$; the dashed arrows $\sigma(f)$ and $E(\rho)(f)$ are just plain functions, not morphisms in $\square_*^*$.

$$\begin{array}{ccccc}
 [1, \dots, 1] & \xrightarrow{x} & \bigvee a & & \\
 \uparrow \sigma(f) & & \downarrow f & \searrow & \\
 [1, \dots, 1] & \xrightarrow{y} & \bigvee b & \longrightarrow & K \\
 & & \uparrow E(\rho)(f) & &
 \end{array}$$

5 The Braiding

This permutation is where our braids come from. But you'll notice that the K side is not involved in the computation of the permutation.

This comes from a rather nice fact that about grothendieck constructions:

$$\begin{array}{ccc}
 \text{Ch}^*(K) = \int (E(\rho) \circ W(K)) & \xrightarrow{Q} & \int E(\rho) = \square^\vee / \rho \\
 \downarrow \pi & \lrcorner & \downarrow \pi \\
 \text{Ch}(K) & \xrightarrow{W(K)} & \square^\vee
 \end{array}$$

The map Q is just a projection to the first arrow of $[1, \dots, 1] \rightarrow \bigvee a \rightarrow K$. And ρ is the classifier for runs. Since E is essentially just the yoneda embedding, the right corner turns into a slice category of wedges to ρ . Secondly, K does not appear on the right hand side.

Lemma 5.1. *The functor Q is always faithful.*

Proof. Let x, y be two objects in $\text{Ch}^*(K)$. Then let $[1, \dots, 1] \rightarrow \bigvee a \rightarrow K$ and $[1, \dots, 1] \rightarrow \bigvee b \rightarrow K$ be two objects in $\text{Ch}^*(K)$. Two morphisms f, g between these objects are equal exactly when they have the same map $\bigvee a \rightarrow \bigvee b$ Which happens if and only if $Q(f) = Q(g)$. So Q is indeed faithful. $\square$

But, we can do a nicer trick

Lemma 5.2. *Let Z be the final precubical set. Then $\text{Ch}^*(Z)$ is isomorphic to $\square^\vee/\rho$.*

Proof. We know Q is faithful already. To show Q is full, again we have two objects $[1, \dots, 1] \rightarrow \bigvee a \rightarrow K$ and $[1, \dots, 1] \rightarrow \bigvee b \rightarrow K$. Let f be a morphism in $\square_*^*/\rho$. That is, $f : \bigvee a \rightarrow \bigvee b$ and it forms a commuting triangle:

$$\begin{array}{ccc} \bigvee a & \longrightarrow & \rho \\ f \downarrow & \nearrow & \\ \bigvee b & & \end{array}$$

But, since Z is the final precubical set, every map commutes in the triangle

$$\begin{array}{ccc} \bigvee a & \longrightarrow & Z \\ f \downarrow & \nearrow & \\ \bigvee b & & \end{array}$$

So Q is full.

Surjectivity on objects is equally straightforward. Every wedge $\bigvee a \rightarrow \rho$ is equivalent to a $[1, \dots, 1] \rightarrow \bigvee a$. And every $\bigvee a$ has a unique $\bigvee a \rightarrow Z$. So we have a corresponding object of $\text{Ch}^*(Z)$. $\square$

So, consider a morphism in the slice $\square_*^*/\rho$. Let $\sigma(f)$ be the corresponding permutation. We need to show that the construction of these permutations is functorial. Define

$$S : \text{Ch}^*(Z) \Rightarrow \sqcup_n \Sigma_n$$

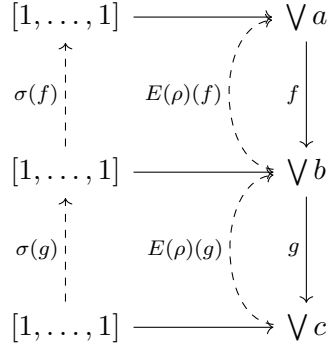
where Σ_n has a single object, and morphisms are permutations of $[n]$ with composition. S sends objects to the object with the length of the wedge, and morphisms $f \mapsto \sigma(f)$.

Lemma 5.3. *The map S is indeed a functor.*

Proof. Firstly, observe that the identity function in $\text{Ch}^*(Z)$ gives us the identity function on runs. Whose corresponding projection is the identity. This induces the identity permutation.

$$\begin{array}{ccc} [1, \dots, 1] & \xrightarrow{x} & \bigvee a \\ \uparrow \text{id} & \curvearrowright \text{id} & \downarrow \text{id} \\ [1, \dots, 1] & \xrightarrow{x} & \bigvee a \end{array}$$

For composition, The diagram below shows we just need $E(\rho)(g \circ f) = E(\rho)(g) \circ E(\rho)(f)$. Which follows from $E(\rho)$ being a functor.



□

Which brings us to the Garside Germ Presentation of braids.

That is, the set $\{[\sigma] \mid \sigma \in \Sigma_n \wedge [\sigma\sigma'] = [\sigma] + [\sigma'] \text{ when } |\sigma\sigma'| = |\sigma| + |\sigma'|\}$ where

$$i(\sigma) := \{(i, j) \in [n] \times [n] \mid i < j \wedge \sigma(j) < \sigma(i)\}$$

and $|\sigma| = |i(\sigma)|$. This is the set inversions. The key fact is a “no double crossings” feature of Ch^* . That is, once you refine a coordinate, you can’t refine it again. Let’s first state this in pure combinatorics

Lemma 5.4. *let f and g be permutations of $[n]$. And suppose that $i(f) \subset i(g \circ f)$. Then $|g \circ f| = |g| + |f|$.*

Proof. Every element of $i(g \circ f)$ either lies in $i(f)$ or not, so

$$i(g \circ f) = (i(g \circ f) \cap i(f)) \sqcup (i(g \circ f) \setminus i(f)).$$

By hypothesis $i(f) \subseteq i(g \circ f)$, so the first set equals $i(f)$ and contributes $|f|$.

For the second, a pair $(k, l) \notin i(f)$ has $f(k) < f(l)$, and then $(k, l) \in i(g \circ f)$ iff $g(f(k)) > g(f(l))$, i.e. iff $(f(k), f(l)) \in i(g)$. Since f is a bijection, $(k, l) \mapsto (f(k), f(l))$ restricts to a bijection

$$i(g \circ f) \setminus i(f) \xrightarrow{\sim} \{(p, q) \in i(g) : f^{-1}(p) < f^{-1}(q)\}.$$

This target is all of $i(g)$: otherwise take $(p, q) \in i(g)$ and set $k = f^{-1}(q) < l = f^{-1}(p)$; then $f(k) = q > p = f(l)$ gives $(k, l) \in i(f)$, while $g(f(k)) = g(q) < g(p) = g(f(l))$ gives $(k, l) \notin i(g \circ f)$, contradicting $i(f) \subseteq i(g \circ f)$. Hence the second set contributes $|g|$, and $|g \circ f| = |f| + |g|$. □

Lemma 5.5. *The map $f \mapsto [S(f)]$ gives us a functor $\text{Ch}^*(Z) \Rightarrow \text{Braid}$, which lifts to $\text{Conc} \Rightarrow \text{Braid}$*

Proof. It always sends objects to the braid object associated with the length of x .

For morphisms, we need to show is that $|S(f \circ g)| = |S(f)| + |S(g)|$ in order for this to be defined.

Let $g : \bigvee a \rightarrow \bigvee b$ and $f : \bigvee b \rightarrow \bigvee c$. According to the previous lemma, it suffices to show that $i(S(f)) \subset i(S(g \circ f))$. Since Ch is monoidal, we can split $\text{Ch}(\bigvee c)$ into $\prod_i (\text{Ch}(\square^{c_i}))$. This splits the triangle of $\bigvee a \rightarrow \bigvee b$ over $\bigvee c$ into one diagram per component. Since *coordMap* is also monoidal, it suffices to prove this for each diagram.

So, we can assume $f : \bigvee b \rightarrow \square^m$. Suppose $(k, l) \in i(f)$. What makes the precubical sets special is that maps from cubes to cubes respects the order embedding. So $\bigvee b \rightarrow \square^m$ must split as $(\bigvee b_l) \vee (\bigvee b_r) \rightarrow \square^m$ such that at the junction vertex, the k coordinate is 1, but the i coordinate is 0. Now, $g : \bigvee a \rightarrow (\bigvee b_l) \vee (\bigvee b_r)$ splits through $\text{Ch}((\bigvee b_l) \vee (\bigvee b_r))$ into $g_l : \bigvee a_l \rightarrow \bigvee b_l$ and $g_r : \bigvee a_r \rightarrow \bigvee b_r$. In other words, $g \circ f = (g_l \circ f_l) \vee (g_r \circ f_r)$ with the same junction point. So $(k, l) \in i(g \circ f)$.

Composition is now well-defined, and in fact $[S(g \circ f)] = [S(g)][S(f)]$. Also $[S(\text{id})] = \text{id}$, so indeed Conc defines a functor. $\square$

ClaudeSlop to review:

Lemma 5.6. *The map $f \mapsto [S(f)]$ is a functor $\text{Ch}^*(Z) \Rightarrow \text{Braid}$, which lifts to Conc .*

Proof. On objects, S sends a wedge to the braid object indexed by its length, and $[S(\text{id})] = \text{id}$. For composition, let $g : \bigvee a \rightarrow \bigvee b$ and $f : \bigvee b \rightarrow \bigvee c$ be composable in Ch . Since $S = \text{coordMap}$ is a functor we have $S(f \circ g) = S(f) \circ S(g)$, so by the previous (“no double crossing”) lemma, applied with $\phi = S(g)$ and $\psi = S(f)$, it suffices to prove

$$i(S(g)) \subseteq i(S(f \circ g)). \quad (*)$$

Write $\bigvee b = \bigvee [b_1, \dots, b_r]$ and identify $\text{coordMap}(\bigvee b) = \coprod_j [b_j]$, ordered first by summand and then within each summand.

Claim 1: every inversion of $S(g)$ has both coordinates in a common summand of $\bigvee b$. Since $\text{coordMap}(g)$ is monotone in its first (summand) coordinate, $k < l$ implies that $S(g)(k)$ and $S(g)(l)$ lie in summands $s_k \leq s_l$ of $\bigvee b$. If $s_k < s_l$, then $S(g)(k)$ lies in a strictly earlier summand than $S(g)(l)$ and hence $S(g)(k) < S(g)(l)$; but (k, l) is an inversion, so $S(g)(k) > S(g)(l)$. Therefore $s_k = s_l$: both coordinates lie in a common summand of $\bigvee b$.

Claim 2: $S(f)$ preserves order inside each summand of $\bigvee b$. By naturality of coordMap , its restriction to the block $[b_j]$ is coordMap of the composite $\square^{b_j} \hookrightarrow \bigvee b \xrightarrow{f} \bigvee c$. This is a map of cubes into a serial wedge, so it lands in a single summand of $\bigvee c$ as a face embedding, and coordMap of a face embedding is an order-preserving injection. Thus for $p < q$ in a common block $[b_j]$ we have $S(f)(p) < S(f)(q)$.

Combining. Let $(k, l) \in i(S(g))$ and set $p = S(g)(l) < q = S(g)(k)$. By Claim 1 the pair p, q lies in a common summand of $\bigvee b$, so by Claim 2, $S(f)(p) < S(f)(q)$. Hence

$$S(f \circ g)(k) = S(f)(q) > S(f)(p) = S(f \circ g)(l),$$

and since $k < l$ this is exactly $(k, l) \in i(S(f \circ g))$, proving $(*)$. Therefore $|S(f \circ g)| = |S(f)| + |S(g)|$, so $[S(f \circ g)] = [S(f)][S(g)]$ and $f \mapsto [S(f)]$ is a functor. $\square$

This looks a bit silly, because all we’ve done is proven our braids words have no cancellation. But that only means we found a nice basis. And in fact, it means we’re working the positive braid monoid. We recover the full braiding structure by lifting to the FreeGroupoid , which is now free.

Lemma 5.7. *For any precubical set K , if there is an $f : \square^2 \rightarrow K$, then $\text{Conc}(K)$ is nontrivial*

Proof. It suffices to find a morphism f in $\text{Ch}^*(K)$ such that $S(Q(f))$ is not the identity. Consider two maps $p : \vee[1, 1] \rightarrow \vee[2]$ going along the bottom edge, then the right edge. And consider the map $q : \vee[1, 1] \rightarrow \vee[2]$ going along the top edge. We have

where $\sigma(p) = \{1 \mapsto 2, 2 \mapsto 1\}$ has an inversion. so $[\sigma(p)]$ is not the identity braid. $\square$

Lemma 5.8. *The category $\text{Conc}(\square^n)$ is isomorphic to the pure braids.*

6 Relationship to the Salvetti Complex

The key fact here is that our story mirrors the salvetti story, on the nose, for cubes. That is, the face complex is exactly $\text{Ch}(\square^n)$, and the salvetti complex is exactly $\text{Ch}^*(\square^n)$. I'd argue this explains why the braid invariant from 'Configuration Spaces' collapsed, and why this complexification survives. However, it does suggest that

1. The salvetti complex is also $\text{Ch}(Z)$, so the quotient by the free action some is also related to the complexification story
2. $\text{Ch}^*(Z)$ is bigger than Braid, so something interesting is happening